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There's a Big Problem With Your Car's Tires

Regulators—and a reluctant industry—are finally starting to deal with it.

BY DAVID ZIPPER

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What image does the phrase *car pollution* bring to mind? You're probably envisioning puffs of exhaust wafting from a vehicle's tailpipe. And for good reason: Gasoline-powered automobiles generate over 16 percent of American greenhouse gas emissions.

But cars—even electric cars—also sully our planet in other ways, like the constant rubbing of tires against pavement. Such friction causes tires to gradually erode, spewing tiny chunks of chemical compounds that float through the air and settle on the ground, where rainwater carries them into streams, creeks, and riverbeds. We are only starting to understand the resulting impact on aquatic ecosystems, but we already know that one tire chemical, called 6PPD, has decimated certain fish populations.

Last week, the Environmental Protection Agency formally acknowledged the ecological damage wreaked by tire pollution. The agency announced that it would investigate the toxicity of 6PPD, a necessary step toward a potential ban on its widespread use in tire manufacturing. According to knowledgeable people I checked with, this is the first time that a federal agency has formally considered tires' ecological harm. At long last, the U.S. is thinking beyond the tailpipe to examine hidden ways that cars befoul the planet.

The scientific community only recently began evaluating tire pollution in earnest, in part because it's tricky to study the impact of individual tire compounds in the real world (far easier to bring a car into a lab, hook up sensors, and measure what comes out of its tailpipe). What we know so far about tire particles, which are a kind of microplastic, is deeply concerning. A 2021 study by the Pew Charitable Trusts found that tire compounds account for more than three-fourths of all microplastics in global oceans, threatening zooplankton that undergird marine food chains.

In a bombshell 2020 study, University of Washington researchers directly tied 6PPD, a chemical in tires intended to prolong the products' life, to the collapse of coho salmon populations in Puget Sound. 6PPD reacts with air to become 6PPD-quinone, a compound so toxic that exposure to even tiny concentrations in water can induce “acute mortality” in fish, according to the study.

“We started with a mix of 2,000 chemicals and were able to get all the way down to this one highly toxic chemical, something that kills large fish quickly,” said Edward Kolodziej, a co-author of the study, when it was first published. “We think it is probably found on every single busy road in the world.”

In a 2022 report, the California Department of Toxic Substances Control noted that the extirpation of coho salmon from the San Francisco Bay Area coincided with 6PPD's widespread adoption during the latter half of the 20th century. This past June, that California department required tire manufacturers to search for a compound that could replace 6PPD. Recent research has found 6PPD-quinone to be similarly lethal to other fish species, including brook trout, rainbow trout, chinook salmon, and white-spotted char.

A national effort to mitigate 6PPD's damage requires federal action. That puts the spotlight on the EPA, which has authority under the [Toxic Substances Control Act](#) to ban toxic chemicals. As the alarm bells began to sound in the West, the EPA created a [6PPD resource page](#) on its website and last year launched a cross-agency working group to coordinate its response.

In August, the Yurok, the Port Gamble S'Klallam, and the Puyallup tribes (all of which are based in the Pacific Northwest, where they have long harvested salmon) [submitted a petition](#) to the EPA requesting a prohibition on 6PPD, citing 6PPD-quinone's toxicity for fish as well as its potential harm to humans.* In October, the attorneys general of Connecticut, Oregon, Rhode Island, Vermont, and Washington [sent a letter](#) of support.

“What’s startling and frightening about 6PPD-quinone is that it’s a ubiquitous chemical now,” said Elizabeth Forsyth, a senior attorney at Earthjustice who worked with the tribes on their EPA petition. “The fact that it’s toxic to coho salmon is one thing, but is it toxic to people in Maine or wildlife in the Amazon? We don’t know, but it’s likely.”

Last week, the EPA [granted the tribes’ petition](#), launching a process that could lead to an outright ban on 6PPD. The EPA’s letter did not offer a time frame for reaching a decision, but Forsyth said she expects the process to take several years.

Unsurprisingly, the tire industry is following the 6PPD story closely. In May, [Reuters reported](#) that Michelin, Continental, and Pirelli are searching for chemicals to replace 6PPD. But the industry’s lobbying arm, the U.S. Tire Manufacturers Association, does not appear to be in a hurry. Its [6PPD webpage claims](#), “There are still many questions that need to be answered in order to drive solutions” to the risks of 6PPD. After the EPA announced its investigation last week, the tire association [issued a statement](#) promising “to work closely with the EPA,” but also warned that “any premature prohibition on the use of 6PPD in tires would be detrimental to public safety and the national economy.” The association did not respond when I asked what an appropriate time frame for eliminating 6PPD would be.

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Should the EPA prohibit 6PPD, the environmental benefits could accrue quickly. Since most car owners replace their tires within five years, virtually all tires constructed with 6PPD could be retired within a decade of a ban. But that doesn't mean that the hazards of tire pollution will have abated. New chemicals could take the place of 6PPD, and then there are the hundreds of other, currently used ones whose individual and collective environmental impacts remain largely unknown. As Susanne Brander, an ecotoxicologist at Oregon State University, recently told the Associated Press, "My concern is we're really focused on this one chemical but in the end, it's the mixture. It's many different chemicals that fish are being exposed to simultaneously that are concerning."

It's not just fish we should be worried about. Toxins accumulate up the food chain, meaning that creatures eating a lot of fish—including people—could be at elevated risk. The EPA's 6PPD response cited "limited data to inform a human health risk assessment," but a white paper issued earlier this year by Imperial College London suggested that tire chemicals may be linked to ailments including kidney damage, birth defects, and respiratory problems. The European Union is poised to issue its first tire regulations as part of Euro 7, an upcoming package of vehicle emission rules, but no similar move is on the horizon in the U.S.

Tire pollution is such a nascent research field that we are only starting to understand its impact on wildlife and human health. Worryingly, such dangers could rise if consumers shift toward heavier, overpowered electric vehicles that shred tires faster than equivalent gas-powered models—as automakers seem to envision. Already, J.D. Power found that rapid tire wear is EV owners' top concern.

Tire manufacturers, which pour billions of dollars into research and development, will hopefully create new compounds that mitigate the damage caused by their products. But another option is so obvious that it's easy to miss: Drive less—and ride transit, trains, and bicycles more. The same holds true for other automotive harms that are easily overlooked, including obstructed animal migration routes and potentially toxic roadside dust from eroded asphalt and brake pads. Regulating one chemical at a time is important. But shredding less rubber would be even better. ■

Future Tense is a partnership of Slate, New America, and Arizona State University that examines emerging technologies, public policy, and society.

Correction, Nov. 13, 2023: This article originally described several tribes as being based in Washington state. In fact, one of them, the Yurok, is based in Northern California.

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